

Optimization methods

Sample solutions and their values for instances considered in the starting task

March 24, 2026

Solution	Value
tai20 5 0	
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)	18286
(20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1)	18752
(1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20)	17132
(2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19)	19100
tai20 10 0	
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)	26671
(20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1)	27678
(1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20)	26457
(2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19)	25156
tai20 20 0	
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)	40249
(20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1)	40608
(1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20)	38684
(2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19)	42249
tai100 10 0	
(1, 2, 3, ..., 98, 99, 100)	387110
(100, 99, 98, ..., 3, 2, 1)	386475
(1, 3, 5, ..., 95, 97, 99, 2, 4, 6, ..., 96, 98, 100)	386326
(2, 4, 6, ..., 96, 98, 100, 1, 3, 5, ..., 95, 97, 99)	394767
tai100 20 0	
(1, 2, 3, ..., 98, 99, 100)	464903
(100, 99, 98, ..., 3, 2, 1)	473833
(1, 3, 5, ..., 95, 97, 99, 2, 4, 6, ..., 96, 98, 100)	466170
(2, 4, 6, ..., 96, 98, 100, 1, 3, 5, ..., 95, 97, 99)	462968
tai500 20 0	
(1, 2, 3, ..., 498, 499, 500)	8147610
(500, 499, 498, ..., 3, 2, 1)	8096620
(1, 3, 5, ..., 495, 497, 499, 2, 4, 6, ..., 496, 498, 500)	8188450
(2, 4, 6, ..., 496, 498, 500, 1, 3, 5, ..., 495, 497, 499)	8074884